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2024249

**BE I st Year Examination, Jan 2024**
**BE I[st] Year (CS A&B)**
**1ETRC4: Basic Electronics**

**Duration: 3 hours** **Max marks: 60**

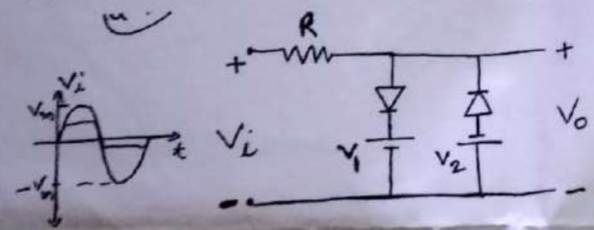
**NOTE: Attempt any Two parts out of Three in each question. Take suitable assumption where needed.**

**Q1) A.** Describe conduction in a p-n junction diode. Also explain its characteristics. (6)

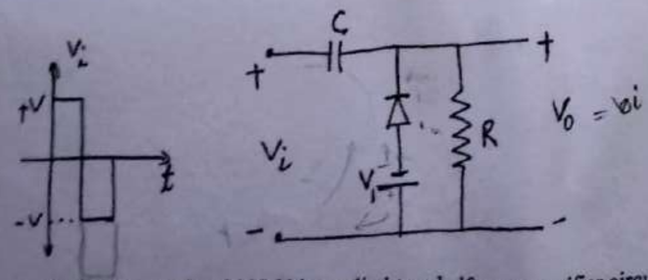
**B.** Compare between insulator, conductor and semiconductor. (6)

**C.** When a reverse bias is applied to germanium p-n junction diode, the reverse saturation current at room temperature is 0.3μA. Determine the current flowing in the diode when 0.15V forward bias is applied at room temperature. (6)

**Q2) A.** Draw the output waveform for the following clipper circuit. (6)




**B.** Draw the output waveform of the clamper circuit. (6)




**C.** An a.c. supply of 230 V is applied to a half-wave rectifier circuit through a transformer of turn ratio 10: 1. Find (i) the output d.c. voltage and (ii) the peak inverse voltage. Assume the diode to be ideal. (6)

**Q3) A.** Design a voltage regulator that will maintain an output voltage of 20 V across a 1-kΩ load with an input that will vary between 30V and 50V. That is, determine the proper value of $R_S$ and the maximum current $I_{ZM}$. (6)

**B.** Write short notes on the following. (3×2)

i) Varactor diode ii) Tunnel Diode

**C.** Explain the working of Zener diode as voltage regulator with all three cases. (6)

**Q4) A.** Compare between CB, CE and CC configuration of an NPN transistor. (6)